

## LESSON 1: FROM DATA TO INFORMATION

Most of our interaction with the world relies on our ability to form an understanding of it; we need to know about it. As we have previously learned, our interactions with the world around us have three basic phases: sensing, processing, and responding. The sensing phase provides us with very basic signals from our environment: light, sound, touch, smell, and others. These are merely signals that we are capable of acquiring and converting into electrical signals our brain can use. It is only after we process these signals that they become meaningful; they become information. Prior to that, they were merely data.

### EXAMPLE 1.1

When we see something, it is initially light signals perceived as various colors and shapes. But once we process them, these colors and shapes become our best friend's face. The colors and shapes in the light signals now have meaning; the data was transformed into information.

**DATA IS ANY FORM OF RAW INPUT THAT CAN BE PROCESSED INTO MEANINGFUL INFORMATION.**

In this definition of 'data', the term 'meaningful' is added to emphasize that processing imparts meaning to the acquired data, thereby converting it into information. It is meaning that differentiates data from information. The same applies to computers. Regardless of whether a human or a computer is processing data, the act of processing is what converts data into information.

### EXAMPLE 1.2

A sensor inputs the following stream of numbers into a computer: '29, 28, 33, 37, 38, 39, 38, 31, 29, 28'. As a list of numbers, this data is meaningless. Let's begin to process it.

1. Identify the sensor that captured the data: Thermometer.
2. Label the data according to the sensor that captured it: Temperature (degrees Celsius).

So far, we have determined that this list of numbers is actually a list of temperatures. One would then want to know, what are these temperatures of?

3. Location of sensor: Outdoor.

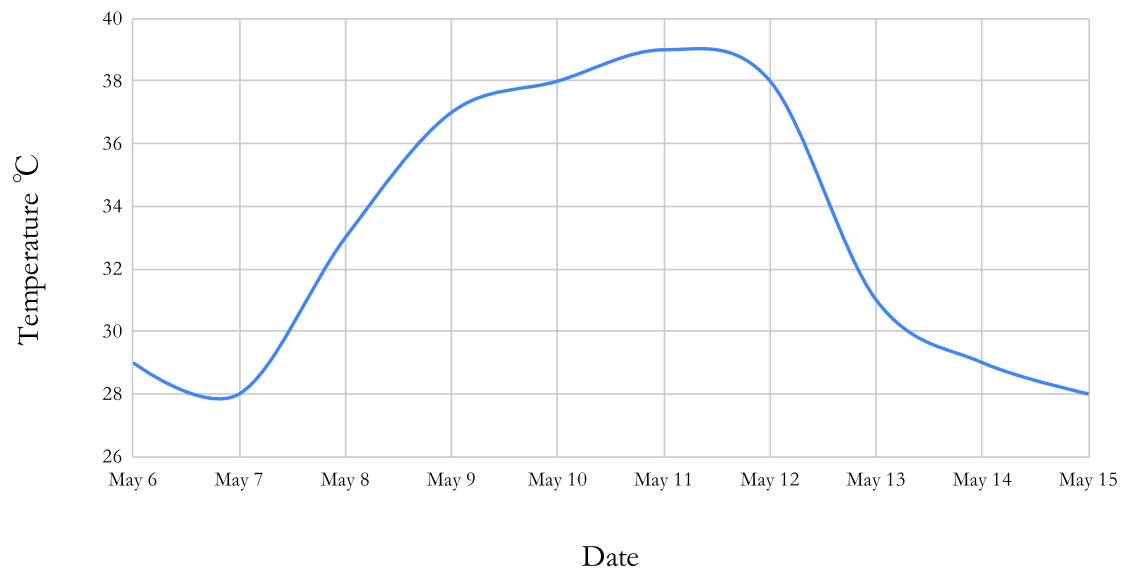
Now that we have gained more information, we know that these numbers are the outdoor temperatures. But when were these readings obtained? Typically, sensors would keep a date and time stamp with each reading. As part of our processing of this data, let us transform this list of temperature data into a table that contains the date and time each reading was taken.

4. Compile date, time and temperature data into a table.

| Date    | May 6 | May 7 | May 8 | May 9 | May 10 | May 11 | May 12 | May 13 | May 14 | May 15 |
|---------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|
| Time    | 12:00 | 12:00 | 12:00 | 12:00 | 12:00  | 12:00  | 12:00  | 12:00  | 12:00  | 12:00  |
| Temp °C | 29    | 28    | 33    | 37    | 38     | 39     | 38     | 31     | 29     | 28     |

Now we have realized that processing data by compiling it with other related data, in this case the date and time of each temperature reading, adds more information. We also learned that organizing data and presenting it in tabular form allows us to better visualize and understand the data. So, the presentation of data, which requires processing, is an added layer of information. Another method of presenting or visualizing data is charting.

### Outdoor Temperature Reading in May



So we have gone from pure raw data of numbers to information; we have learned that between May 8 and May 12, there was a sudden rise and fall in temperature in the area from which the data was collected. This leads us to one last step, interpreting the data. A sudden rise in temperature could be interpreted as a heat wave, and therefore the information learned from our data is that: 'between May 8 and May 12, a heat wave occurred in the area from which the data was collected.' This information was derived through the processing of data, which included identifying, organizing, visualizing, and interpreting the data.

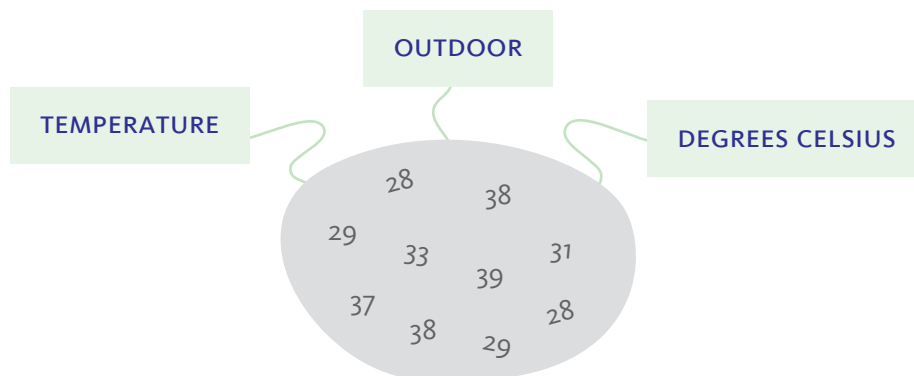
CONVERTING DATA TO INFORMATION INCLUDES IDENTIFYING, ORGANIZING, VISUALIZING, AND INTERPRETING THE DATA.

## IDENTIFYING DATA

One of the initial steps in processing data is identifying what it represents. The information extracted from data is significantly influenced by how it is identified. In the previous example, the list of numbers: `29, 28, 33, 37, 38, 39, 38, 31, 29, 28`, could represent the number of students going on a field trip, or the number of views someone received on their YouTube posting. Therefore, identifying the data by determining its source and what it represents is critical, as it sets the foundation for all subsequent steps.

## LABELING DATA

Labeling data is an essential part of identifying it. This helps prevent mistakes and ensures that everyone examining the data understands its context and meaning. So in the case of Example 1.2, the labels that would be required are: 'temperature', 'outdoor', and 'degrees Celsius'.



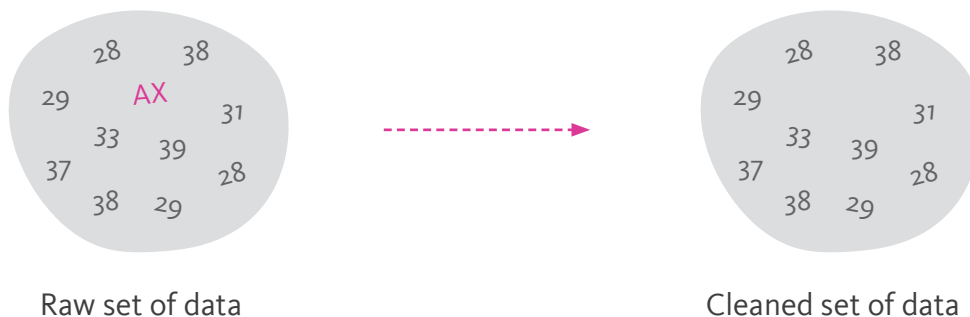
## ORGANIZING DATA

Another important step in the process of extracting information is organizing the available data. There are several ways in which data can be organized: cleaning, structuring, and sorting.

## CLEANING DATA

Cleaning data is the process of removing erroneous data. Errors can enter a dataset for several reasons: errors during input, such as a faulty sensor in an automated system or a mistake by a person entering the data manually. Other errors can arise while data is being stored, waiting to be processed. While there are other reasons errors can occur, the important point is that data should be screened and errors removed.

For example, what if our list of numbers read: `29, 28, 33, AX, 37, 38, 39, 38, 31, 29, 28` or `29, 28, 33, 300, 37, 38, 39, 38, 31, 29, 28`? In either case, the data items 'AX' and '300' do not fit for obvious reasons: an outdoor temperature must be a number, not letters, and 300°C is certainly too high a temperature to be recorded outdoors on Earth, unless maybe you are near a volcano or on a different planet, but that is not the situation.



## STRUCTURING DATA

Data is generally stored in computers in a certain manner that determines how each item of data relates to the others. These are called data structures. We will explore them in greater detail later in this chapter, but for now, we will simply identify two data structures: lists and tables.

1. **List (Array):** An arrangement of data in a linear fashion; all data items are positioned next to each other in a line.

- 2. Table:** Well-suited to organize multiple sets of data. In Example 1.2, the initial organization of the temperature data was linear, with all the numbers presented in an array. But when other sets of data were included, such as the date and time, a table was used to organize the data.

Data structured as an array with one item following the other.

29, 28, 33, 37, 38, 39, 38, 31, 29, 28

Data structured as a table with rows and columns.

| Date    | May 6 | May 7 | May 8 | May 9 | May 10 | May 11 | May 12 | May 13 | May 14 | May 15 |
|---------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|
| Temp °C | 29    | 28    | 33    | 37    | 38     | 39     | 38     | 31     | 29     | 28     |

## SORTING DATA

Sorting plays an important role in data organization and processing. It is the process through which data is rearranged based on certain criteria, such as alphabetical order for alphanumeric data, increasing or decreasing value for numeric data, or by category for data that can be categorized. In the previous example, the data was already sorted. The sorting criterion was date, meaning that the temperature data was in chronological order based on the date it was acquired.

Data sorted chronologically by increasing dates.

| Date    | May 6 | May 7 | May 8 | May 9 | May 10 | May 11 | May 12 | May 13 | May 14 | May 15 |
|---------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|
| Temp °C | 29    | 28    | 33    | 37    | 38     | 39     | 38     | 31     | 29     | 28     |

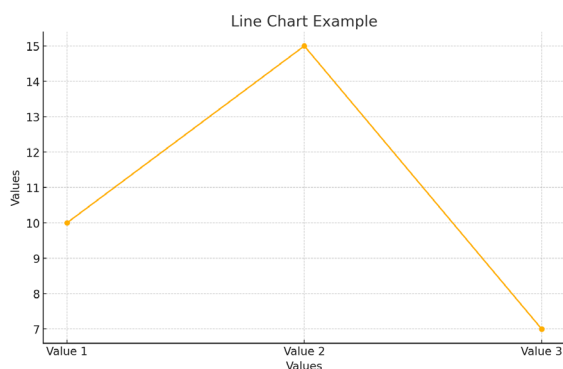
Data sorted by increasing temperature readings.

| Date    | May 7 | May 15 | May 6 | May 14 | May 13 | May 8 | May 9 | May 10 | May 12 | May 11 |
|---------|-------|--------|-------|--------|--------|-------|-------|--------|--------|--------|
| Temp °C | 28    | 28     | 29    | 29     | 31     | 33    | 37    | 38     | 38     | 39     |

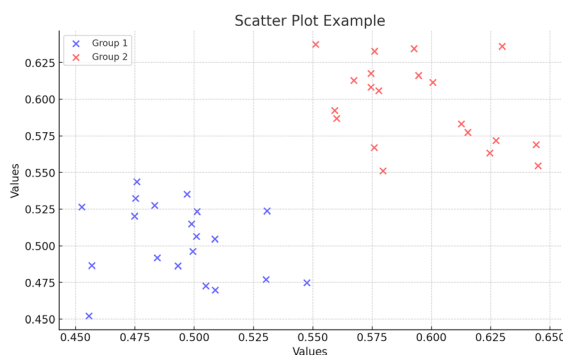
Each method of sorting allows us to better identify certain aspects of the data. Data sorted chronologically by increasing dates allows us to more easily identify trends over time, while data sorted by increasing temperature readings allows us to more easily identify the lowest and highest temperature readings collected in the dataset.

## VISUALIZING DATA

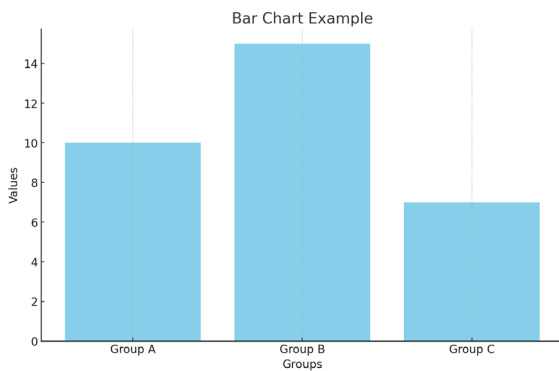
To visualize data means to present it in a way that makes it easy to visually identify trends and patterns. A common and effective method to visually present data is through charts. There are different types of charts, each presenting the data in a slightly different way. Four common types of charts are: line chart, scatter plot, bar chart, and pie chart.



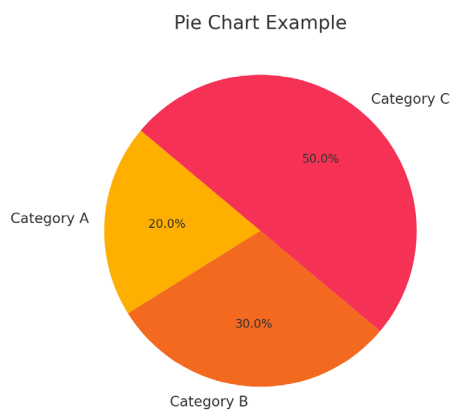
A **line chart** is a chart in which the value of each data point is represented by a point on the chart, and the points are then connected by a line. Line charts are useful for showing trends over time or continuous data.



A **scatter plot** is similar to a line chart in that the value of each data point is represented by a point on the chart. However, in a scatter plot, the points are not connected by a line. Instead, scatter plots are used to display the relationship between two continuous variables, making it easier to identify patterns, trends, and correlations.



A **bar chart** is a chart in which data is organized into groups or categories, with the value of each category represented by a bar. Bar charts can be either vertical or horizontal and are useful for comparing quantities across different categories.



A **pie chart** is a chart in which data is organized into groups or categories that are part of a whole. Each category is represented by a slice of the pie, which indicates the proportion of the whole that this category represents. Pie charts are useful for showing relative proportions and percentages of a whole.

The choice of the type of chart used to visualize data depends on the type of data itself. Generally, data can be broadly classified into two main types: quantitative and qualitative. Such classification influences the handling and presentation of the data.

## Notes

Understanding the transformation of data into information is a crucial skill that extends beyond computer science, impacting fields like healthcare, business, finance, education, and environmental studies. This fosters critical thinking and analytical skills, enhancing our ability to interpret and act on complex data.

## Quantitative Data

Quantitative data is numerical data that can be measured or counted, which means that it can either be continuous or discrete. Continuous data is essentially measurable data that can take any value within a range, such as temperature, height, and time. This type of data is best visualized using line charts and scatter plots. For example, in Example 1.2, a line chart was used to visually present the data because line charts are best suited for showing trends over time or continuous progression, such as the change in temperature over time.

**QUANTITATIVE DATA IS NUMERICAL DATA THAT CAN BE MEASURED OR COUNTED.**

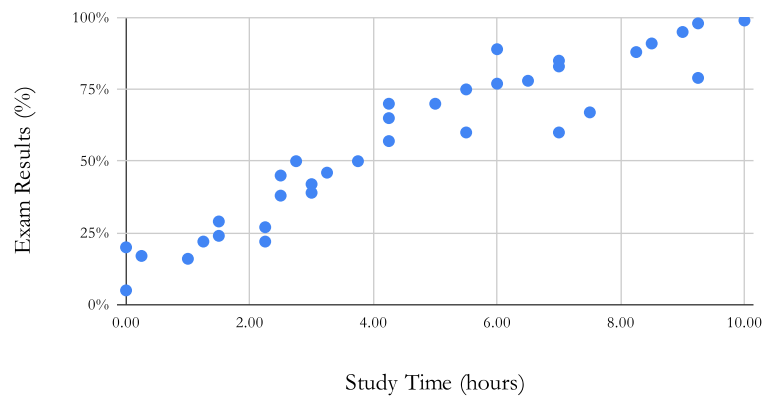
### EXAMPLE 1.3

Data was collected from 35 students. Two sets of data were obtained: the hours of study for an exam and the results achieved on that exam. This is the raw data unprocessed in the form of a list, or more specifically, a paired list:

(3.00, 42%), (5.00, 70%), (7.00, 60%), (1.25, 22%), (2.50, 38%), (10.00, 99%), (6.00, 77%), (0.00, 20%), (8.50, 91%), (4.25, 70%), (0.00, 5%), (1.50, 24%), (6.50, 78%), (0.25, 17%), (2.75, 50%), (9.00, 95%), (9.25, 98%), (1.00, 16%), (2.25, 27%), (7.50, 67%), (9.25, 79%), (1.50, 29%), (2.50, 45%), (2.25, 22%), (7.00, 83%), (3.25, 46%), (6.00, 89%), (4.25, 65%), (3.00, 39%), (5.50, 75%), (3.75, 50%), (8.25, 88%), (4.25, 57%), (5.50, 60%), (7.00, 85%)

Looking at this data, it is unclear what information it holds. However, if it is processed and charted in the form of a scatter plot, the information it holds becomes better visualized.

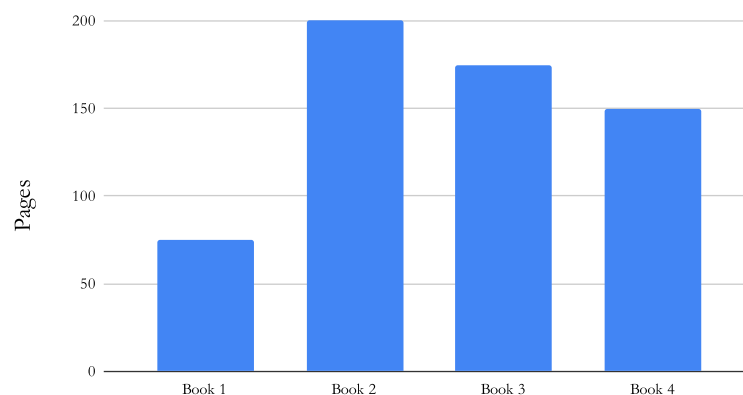
Exam Results (%) vs. Study Time (hours)



The scatter plot of exam results versus study time reveals a pattern: as the hours of studying increase, so do the exam results. While it is not a perfect straight line, it does reveal a trend that is not noticeable in the raw data.

Discrete data, on the other hand, is typically countable data that can only take distinct values, such as the number of pages in a book or the number of students in a class. It is not continuous, meaning it cannot be in the form of fractions or decimals. For example, you cannot have  $10\frac{1}{2}$  pages in a book or 20.25 students in a class. The values of discrete data can only be whole numbers. In general, bar charts are a good choice for visually representing discrete data.

Number of Pages in Books



The bar chart allows for comparison of the number of pages in each book.

## Qualitative Data

Qualitative data is descriptive data that can be observed but not measured. It is also referred to as categorical data because it can be categorized into groups and labeled, such as types of fruits, names of cities, colors of cars, and yes/no responses in a survey. This type of data is best visualized using bar charts and pie charts, which allow for a clear representation of categories and their frequencies. However, before it can be represented in charts, this type of data must be converted into quantitative data. The most common method of making this conversion is by counting the data items that fall within each category.

**QUALITATIVE DATA IS DESCRIPTIVE DATA THAT CAN BE OBSERVED BUT NOT MEASURED.**

### EXAMPLE 1.4

The school's information management system is backed by a database that contains data about students attending the school, including their names, ages, grades, gender, and other pertinent information. The school has 618 students enrolled in Grades 1 through 12, which is too many to show in this book. Here are the first 6 entries in the database table:

| Name    | Family  | Age | Grade | Gender |
|---------|---------|-----|-------|--------|
| Dana    | Lankart | 6   | 1     | Girl   |
| Ahmed   | Dandash | 6   | 1     | Boy    |
| Ruba    | Najem   | 8   | 3     | Girl   |
| Tarek   | Najem   | 10  | 5     | Boy    |
| Thomas  | Johnson | 9   | 4     | Boy    |
| Ghassan | Mekki   | 9   | 4     | Boy    |

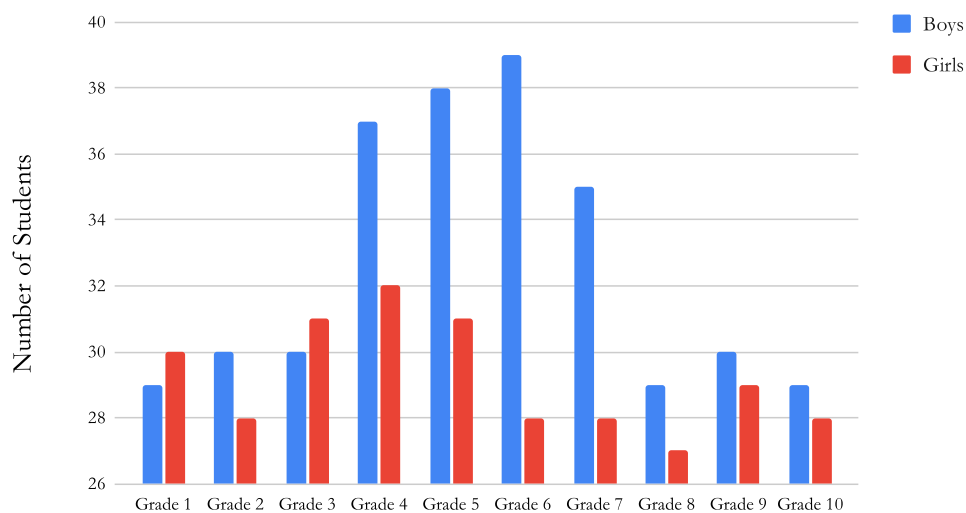
Based on this table, grade levels and gender are considered categorical qualitative data. The data items of grade levels are the names of each grade, such as 'Grade 1', 'Grade 2', and so on up to 'Grade 12'. The data items under gender are categorized as 'Boy' and 'Girl'. None of these data items are measurable or numerical in nature.

Assuming we want to learn the number of boys and girls in each grade level, the first step would be to query the database to count the number of occurrences of the data items 'Boy' and 'Girl' for each grade. This query results in the conversion of qualitative data into discrete quantitative data, from which we can extract the following information:

|       | Grade 1 | Grade 2 | Grade 3 | Grade 4 | Grade 5 | Grade 6 | Grade 7 | Grade 8 | Grade 9 | Grade 10 |
|-------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| Boys  | 29      | 30      | 30      | 37      | 38      | 39      | 35      | 29      | 30      | 29       |
| Girls | 30      | 28      | 31      | 32      | 31      | 28      | 28      | 27      | 29      | 28       |

Since the data can be grouped by grade levels (Grade 1 - Grade 10) and gender (boy, girl) it is better visualized using a bar chart. The resulting bar chart would look like this:

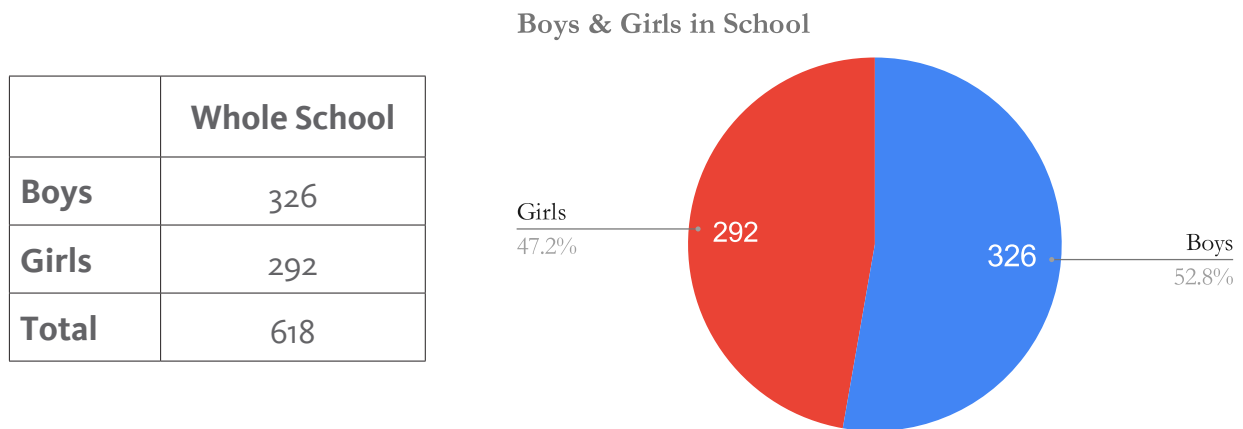
**Boys & Girls in Grade Levels**



The bar chart reveals that the number of boys exceeds the number of girls, particularly in grades 4, 5, 6, and 7.

## EXAMPLE 1.5

If, for the same data, we wanted to visualize the proportion of boys and girls in the whole school, we could use a pie chart. Since we are now interested in the entire school rather than the details of each grade level, a pie chart is appropriate. To obtain this information, we would add the number of boys across all grades and the number of girls across all grades. The two categories, ‘Boys’ and ‘Girls,’ can then be represented in a pie chart, showing that the number of boys and girls is nearly the same.



## INTERPRETING DATA

The final step in the process of converting raw data to actionable information is the interpretation of the findings. This step involves explaining what the data is presenting to us, whether it is a spike in outdoor temperature, a relationship between two sets of data such as study time and exam results, or a discrepancy between the proportion of boys and girls in a school. Interpreting these findings poses two crucial questions: “Why?” and “What should we do?”

The first question, “Why?” forces us to identify the factors underlying our findings. This step is essential for uncovering the reasons that have led to the

information we have discovered. It adds an extra layer of understanding that is necessary for a deeper comprehension of the world around us.

The second question, “What should we do?” prompts us to consider our next steps based on our findings. These next steps may involve further investigating the underlying causes or, if we are satisfied with our findings, learning from them and incorporating them into our decision-making process. This kind of information is referred to as ‘actionable information’, which is information that we can use to make informed decisions.

The process of converting raw data into information is fundamental to understanding and interpreting the world around us. By carefully following the basic stages of identifying, organizing, visualizing, and interpreting data, we can transform unprocessed data into actionable information regardless of the initial nature of the data, whether it is quantitative or qualitative. This process should guide our learning and decision-making, providing us with the analytical tools necessary to make informed decisions based on accurate and meaningful information.

### Career Tips

Understanding the process of converting raw data into actionable information can open doors to a variety of rewarding careers, such as data analysts, data scientists, business intelligence developers, financial analysts, and market research analysts. All of these roles are in high demand in today’s data-driven world.

## SUMMARY

The essential steps and processes for converting raw data into information are categorized into four main stages:

1. Identifying Data: This includes collecting, identifying, and labeling data.
2. Organizing Data: This includes cleaning, structuring, and sorting data.
3. Visualizing Data: Creating visual representations (charts, graphs) to highlight trends and patterns.
4. Interpreting Data: Analyzing visualized data to extract meaningful insights that lead to informed decision-making.

Data can be categorized as quantitative and qualitative.

Quantitative data is numerical data that can be measured (continuous) or counted (discrete).

Qualitative data is descriptive data that can be observed but not measured and is referred to as categorical data because it can be categorized into groups and labeled.

Four common charts used to visualize data are line charts, scatter plots, bar charts, and pie charts. Line charts and scatter plots are best for continuous quantitative data, while bar charts and pie charts are best for discrete and categorical data.

### Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

| T/F      | STATEMENT                                                                              |
|----------|----------------------------------------------------------------------------------------|
| <u>F</u> | The process of converting raw data into information only includes organizing the data. |
| <u>T</u> | Labeling data helps prevent mistakes.                                                  |
| <u>F</u> | Continuous data can only take discrete values.                                         |
| <u>T</u> | A line chart is best for showing trends over time.                                     |
| <u>F</u> | Categorical data represents measurements that can take any value within a range.       |
| <u>T</u> | A scatter plot is best suited for displaying continuous data.                          |
| <u>T</u> | Cleaning data involves removing erroneous data.                                        |
| <u>F</u> | A pie chart is suitable for visualizing continuous data.                               |
| <u>T</u> | Sorting data arranges it for easy access and analysis.                                 |
| <u>T</u> | Interpreting data involves explaining what the data is presenting.                     |

Activity 1.2

Answer the questions below.

1. What is the first step in converting raw data into information?
  - a) Visualizing Data
  - b) Organizing Data
  - c) **Identifying Data**
  - d) Interpreting Data
  
2. Which of the following charts is best used for showing trends over time?
  - a) Bar Chart
  - b) **Line Chart**
  - c) Pie Chart
  - d) Scatter Plot
  
3. What is the primary purpose of cleaning data?
  - a) To identify data
  - b) To create charts
  - c) **To remove erroneous data**
  - d) To label data
  
4. Which type of data can take any value within a range?
  - a) Categorical Data
  - b) Discrete Data
  - c) **Continuous Data**
  - d) Nominal Data

5. What does a scatter plot best represent?
  - a) Proportions within a whole
  - b) Trends over time
  - c) Relationships between two continuous variables
  - d) Comparisons across categories
  
6. What type of chart would you use to compare quantities across different categories?
  - a) Bar Chart
  - b) Line Chart
  - c) Pie Chart
  - d) Scatter Plot
  
7. What type of chart is best for showing proportions of a whole?
  - a) Bar Chart
  - b) Line Chart
  - c) Pie Chart
  - d) Scatter Plot
  
8. Which of the following is a key aspect of data visualization?
  - a) Removing erroneous data
  - b) Arranging data in a list
  - c) Presenting data in a visual format
  - d) Labeling data points
  
9. Why is it important to interpret data after visualizing it?
  - a) To ensure data is labeled correctly
  - b) To derive meaningful insights and make decisions
  - c) To remove duplicate entries
  - d) To organize data in a table

### Activity 1.3

Read the provided data examples. Classify each example as either quantitative or qualitative.

#### DATA EXAMPLES

1. *Types of Pets: Different kinds of pets owned by students.*
2. *Temperature: Daily temperatures recorded over a month.*
3. *Weight: The weight of different fruits*
4. *Brands of Cars: Different car brands owned by people.*
5. *Time: The time taken by athletes to complete a race.*
6. *Blood Types: Blood types of students.*

Quantitative : 2. Temperature, 3. Weight, 5. Time

Qualitative : 1. Types of Pets, 4. Brands of Cars, 6. Blood types

### Activity 1.4

Identify 2 examples of quantitative data and 2 examples of qualitative data from your daily life.

Quantitative : Measure the time you spend on activities such as studying, exercising, and watching TV.

Qualitative : Record the different modes of transportation you use throughout the week such as bus, car, bike, and walking.

Activity 1.5

Examine the dataset provided in each question and determine the choice that best describes it.

1. List A = {12.5, 12.3, 11.7, 14.6, 15.1}
  - a) Qualitative Categorical Data
  - b) **Quantitative Continuous Data**
  - c) Quantitative Discrete Data
  
2. List B = {red, blue, red, red, yellow, blue, blue, red}
  - a) **Qualitative Categorical Data**
  - b) Quantitative Continuous Data
  - c) Quantitative Discrete Data
  
3. List C = {yes, yes, no, no, maybe, maybe, yes, yes, yes}
  - a) **Qualitative Categorical Data**
  - b) Quantitative Continuous Data
  - c) Quantitative Discrete Data
  
4. List D = {2, 3, 7, 2, 4, 1, 1, 1, 4, 5}
  - a) Qualitative Categorical Data
  - b) Quantitative Continuous Data
  - c) **Quantitative Discrete Data**
  
5. List E = { $\frac{1}{2}$ ,  $8\frac{1}{2}$ , 7,  $8\frac{1}{4}$ ,  $5\frac{3}{4}$ }
  - a) Qualitative Categorical Data
  - b) **Quantitative Continuous Data**
  - c) Quantitative Discrete Data

## Activity 1.6

Examine the datasets provided and read the supporting narrative. Based on the information given, use a spreadsheet application to generate a chart that visually represents the data. Ensure that the data is appropriately organized, and that both your tabulated data and chart are clearly labeled. Identify patterns and draw conclusions from the data.

Refer to the spreadsheet '[IT Frontiers A Part 2 - Lesson 1 - Activity 1.6](#)'.

1. Age\_Height\_List = { (11, 143), (12, 145), (13, 153), (14, 162), (15, 170), (16, 175), (17, 178), (18, 180), (19, 181), (20, 181) }

Format: (age, height)

Omar's pediatrician has been keeping track of his height from the age of 11 to 20 years. Every year, she records his height in centimeters.

2. Rainfall\_Weight\_List = { (12, 150), (10, 120), (20, 180), (11, 140), (15, 160), (13, 137), (19, 195), (10, 135), (18, 180), (11, 115), (17, 180), (14, 142), (13, 155), (13, 130), (17, 165), (11, 134), (14, 158), (12, 141), (16, 180), (16, 162) }

Format: (rainfall, weight)

A farmer records the weekly rainfall in millimeters and the weight of the fruit picked from the tree in grams. The weekly rainfall ranged between 10 and 20 mm.

3. Item\_Count\_List = { (Laptops, 200), (Smartphones, 300), (Tablets, 170), (Desktops, 150) }

Format: (item, count)

An electronics store manager keeps a record of the items in his store. He wants to present this data to his administration. He queries the database for a count of certain items and receives this response.

4. Coats\_Beachwear\_List = { (January, 220, 50), (February, 200, 60), (March, 170, 100), (April, 130, 150), (May, 100, 200), (June, 50, 250), (July, 40, 230), (August, 30, 210) }

Format: (month, winter coats, beachwear)

A fashion retailer wants to compare the sales of winter coats and beachwear over the last 8 months.

### Notes

A list can contain items in the form of doubles, triplets, or more, which are referred to as tuples. Tuples are similar to lists, but the items in a tuple have a fixed order and cannot be changed.

## Activity 1.7

Examine the provided datasets and supporting narrative. Using the information given, use a spreadsheet application to generate a chart that visually represents the data. Organize the data appropriately, ensuring that both your tabulated data and chart are clearly labeled. Identify and clean any erroneous data in the dataset. Finally, identify patterns in the data and draw conclusions. Refer to the spreadsheet '[IT Frontiers A Part 2 - Lesson 1 - Activity 1.7](#)'.

1. Climate\_Data\_List = { (20th, 310, 14.5), (19th, 290, 14.0), (18th, 280, 13.8), (17th, 275, 13.6), (16th, 270, 13.5), (15th, 265, 13.3), (14th, 260, 13.2), (13th, 255, 13.1), (12th, 250, 13.0), (11th, 245, 12.9), (10th, 240, 12.8), (9th, 235, 12.7), (8th, 230, 12.6), (7th, 225, 12.5), (6th, 220, 12.4), (5th, 215, 12.3), (4th, 210, 12.2), (3rd, 205, 12.1), (2nd, 200, 12.0), (1st, 195, 11.9) }

Format: (century, CO<sub>2</sub> level, temperature)

Sarah is a climatologist. She is studying global warming and is collecting ice core samples from Antarctica to measure the carbon dioxide levels trapped in the ice for each century. She is also measuring the ratios of gases trapped in the ice samples to estimate global temperatures at the time the ice was formed.

2. Coral\_Reef\_List = { (25.0, 8.2), (24.8, 8.3), (25.2, 8.1), (24.5, 8.5), (26.0, 7.8), (24.3, 8.7), (24.4, 8.6), (25.0, 8.0), (24.2, 8.8), (25.5, 8.0), (24.5, 8.4), (26.2, 7.5), (25.8, 7.6), (24.1, 8.9), (24.5, 19.1), (25.2, 7.9), (24.9, 8.1), (26.5, 7.4), (24.7, 8.5), (24.3, 8.6), (26.0, 7.7), (24.8, 8.3), (24.5, 8.5), (25.1, 8.0), (26.3, 7.5), (24.0, 9.0) }

Format: (temperature, O<sub>2</sub> level)

Jordan is a marine biologist studying the health of coral reefs. He uses underwater sensors to monitor the temperature of the water and the amount of dissolved oxygen at various depths.

3. Movie\_Opinion\_List = { Great, Good, Okay, Great, Good, Bad, Awful, Great, Good, Good, Great, Good, Okay, Great, Good, Bad, 1734, Good, Great, Good, Good, Okay, Bad, Good, Great, Okay, Good, Awful, Great, Good, Great, Good, Great, Good, Okay, Great, Good, Bad, Okay, Good, Great, Okay, Good, Great, Good, Good, Great, Good, Okay, Good, Great, Good, Great, Bad, Good, Great, Good, Good, Great, Okay, Great, Bad, !!!!, Good, Great, Okay, Good, Great, Awful, Good, Great, Good, Great, Okay, Good, Great, Good, Good, Great, Bad, Good, Great, Good, Okay, Great, Good, Good, Great, Good, Bad, Okay, Great }

A movie production company wants to gather audience feedback on their latest film. They categorize the feedback into five levels: Great, Good, Okay, Bad, and Awful. Note that this data is categorical and needs to be converted to discrete quantitative data before it is further processed.